

Name:

I. Given the database schema:

CUSTOMER			PRODUCT		
ID	NAME	City	PID	Pname	Type
1	Brian	Chicago	P1	Cinderella	Books
2	Jane	Houston	P2	Dell XYZ	Computers
3	Katie	Houston	P3	Aladdin	Books
4	John	Houston	P4	HP 123	Computers
5	Leo	San Jose			

ORDER				DETAILS	
O_ID	DATE	CUSTOMER_ID	Amount	O_ID	P_ID
001	May 30	2	200	001	P1
002	June 8	3	500	001	P3
003	June 12	2	100	002	P2
004	May 30	4	300	003	P2
005	June 14	2	400	004	P2
006	May 30	3	300	005	P2
				006	P1

Solve all below queries using **Relational Algebra, SQL**:

1. Find all customers in Chicago.

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2. Find name of customers who don't make any order.

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3. Find name of products which have never been sold.

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4. Find name of customers who have bought all books.

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5. Find name of computers which has been sold in May 30.

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III. R(ABCDEI)

$F = \{A \rightarrow C, AB \rightarrow C, C \rightarrow DI, CD \rightarrow I, EC \rightarrow AB, EI \rightarrow C\}$

1. Find the minimal cover?
2. Find the Normal Forms?